The Bandit-Auction Index of Pavan, Segal and Toikka, Taken Literally

The general theorems stand. The application's index formula does not

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Abstract

Pavan, Segal and Toikka (2014, *Econometrica*) illustrate their theory of dynamic mechanism design with "bandit auctions", in which buyers learn their valuations by consuming. Their equation (15) defines the optimal allocation through an index that places the expectation outside a discounted-reward ratio; the Gittins index that delivers optimality is a ratio of expectations. The difference matters. In a one-buyer example that satisfies all of the paper's assumptions, every incentive-compatible mechanism implementing the printed rule earns at most 0.63133, while a simple incentive-compatible mechanism earns 0.63335. The index also holds today's cost fixed for all future periods, although costs in the model can change over time; with time-varying costs no index rule is optimal in general. Two smaller corrections and a missing condition in Example 5 complete the list. The paper's general results are unaffected.

1 What the paper says

In the bandit auction of Pavan, Segal and Toikka (2014), hereafter PST, a seller allocates one unit per period to one of n buyers or to an outside option. Buyer i's valuation θ_{it} changes only when he is served, a sale to i at date t costs c_{it} (p. 634), and the virtual reward of serving i is $\theta_{it} - c_{it} - 1/\eta_{i0}(\theta_{i0})$, where η_{i0} is the hazard rate of the initial type. PST's equation (15), p. 635, defines bidder i's virtual index as

$$\gamma_{it} = \max_{T} \ \mathbb{E}\left[\frac{\sum_{\tau=t}^{T} \delta^{\tau}\big(\theta_{i\tau} - c_{it} - 1/\eta_{i0}(\theta_{i0})\big)}{\sum_{\tau=t}^{T} \delta^{\tau}}\right],$$

with T a stopping time, and Proposition 2 states that awarding the unit to the highest index, with the outside option at zero, is an optimal mechanism. The first-best index on p. 637 has the same form. The index that makes such a rule optimal, and the one in the references PST cite for the optimality of index policies, is the Gittins index, a ratio of expectations:

$$\gamma^{G}_{it} = \max_{T} \ \frac{\mathbb{E}\big[\sum_{\tau=t}^{T} \delta^{\tau}\big(\theta_{i\tau} - c_{i} - 1/\eta_{i0}(\theta_{i0})\big)\big]}{\mathbb{E}\big[\sum_{\tau=t}^{T} \delta^{\tau}\big]}.$$

The two coincide for fixed stopping dates. They differ when the stopping time depends on what the buyer learns, which is the reason for using a bandit model.

2 The printed index loses revenue

Take one buyer, no cost and $\delta = \frac{1}{2}$. The initial value is $v \sim U(0,1)$, so the virtual reward of a first unit is $r = 2v - 1$. After the first purchase the value moves by $Z \sim N(0,1)$; after the kth purchase, $k \geq 2$, by an independent normal shock with standard deviation $10^{-6}2^{-(k-2)}$. The example satisfies PST's assumptions on the type process (the later shocks only ensure that the innovation distributions are absolutely continuous; their total expected size is $m < 1.6 \times 10^{-6}$).

Under the printed index, the realized discounted average of the rewards from any stopping rule is $r + aZ$ plus later shocks, where the weight a on the first shock lies between 0 and $\frac{1}{2}$ on every path. The index is therefore at most $r + \frac{1}{2}\mathbb{E}Z^{+} + m = 2v - 1 + 0.19947 + m$, which is negative for every $v \leq \frac{2}{5}$, so the printed rule never serves these buyers. The Gittins index is positive as soon as $g(r) \equiv r + \mathbb{E}(r + Z)^{+} > 0$, that is, for $v > 0.36199$. It prices the option to keep buying after a good first draw, which the realized average caps at half its value (Figure 1).

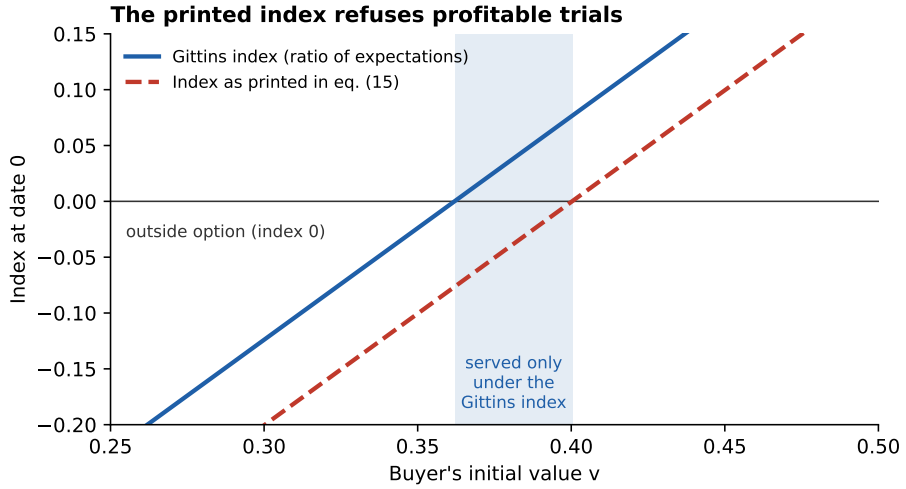


Figure 1: Date-0 index as a function of the buyer's initial value v in the example of Section 2. Blue: the Gittins index. Red, dashed: the index as printed in PST's equation (15). Buyers in the shaded band, $0.362 < v \le 0.400$, are served under the Gittins index and never under the printed one.

Proposition 1. *In this example, every initially individually rational, incentive-compatible mechanism that implements the allocation rule of PST's equations (15) and (16) earns expected revenue of at most* 0.63133. *The following mechanism is globally incentive compatible and individually rational and earns* 0.63335. *A buyer who reports $a \ge 0.36199$ gets one unit at date 0 for $1 - a + \frac{1}{2}g(2a-1)$ and, at date 1, the option to buy every future unit for a single payment of $2(1-a)$; lower reports get nothing.*

Sketch. For the bound, a buyer whom the printed rule never serves contributes no virtual surplus, and a buyer with $v > \frac{2}{5}$ contributes at most $g(2v-1) + m$, the value of one trial followed by the best continuation. Revenue equals expected virtual surplus minus the lowest type's utility, so it is at most $\frac{1}{2}\int_{-1/5}^{1} g(r)\,dr + \frac{3}{5}m = 0.631326$. In the mechanism, a buyer with value y at date 1 takes the option exactly when $y \ge 1 - a$, so a type-v buyer who reports a gets $g(v+a-1) - \frac{1}{2}g(2a-1)$. Since g is convex, $g(v+a-1) \le \frac{1}{2}g(2v-1) + \frac{1}{2}g(2a-1)$, with equality at $a = v$, so truth-telling is optimal and participation is voluntary. Integrating the seller's receipts $\frac{1}{2}[1 + \Phi(2v-1) + \phi(2v-1)]$ over $v \ge 0.36199$ gives 0.633349, which equals the virtual surplus of the Gittins-index rule. The numbers are exact up to the stated digits; we also confirmed incentive compatibility by simulation. □

3 Costs that change over time

PST allow the cost c_{it} to depend on the date, but equation (15) holds today's cost fixed over the whole hypothetical run of the arm. Fixing the formula by putting $c_{i\tau}$ inside the sum does not help, because an arm whose costs change while it waits is no longer a standard (rested) bandit, and the index theorem does not apply. In the same example with $c_0 = 0$ and $c_t = 10$ for $t \ge 1$, a buyer with $v = \frac{3}{8}$ has a positive frozen-cost index (0.0260), so the rule serves him, while any sale at date 0 has expected virtual surplus below -0.2499 under the true costs. Optimality of an index rule requires costs that do not vary with the date.

4 Smaller points

Sign of the handicap. On p. 637 PST describe the optimal mechanism as an efficient one in which the seller's cost is $c_{it} - 1/\eta_{i0}$. The virtual reward $\theta_{it} - c_{it} - 1/\eta_{i0}$ corresponds to the cost $c_{it} + 1/\eta_{i0}$. The displayed objective is right; only the interpretation has the wrong sign.

Means versus absolute moments. PST assume (p. 634) that the innovation distributions have means bounded uniformly in absolute value, and use this (p. 635) to bound expected absolute types. Mean-zero normal shocks with growing variances, for example, have bounded means and unbounded absolute moments. A uniform bound on $\mathbb{E}|\varepsilon_{ik}|$ is sufficient.

Example 5. The main text (p. 632) states that the allocation rule in the nonlinear autoregressive example is implementable whenever the hazard rate is nondecreasing. The Supplemental Material (Section S.2) proves this under two extra conditions, interior solutions and $\phi'' \leq \frac{1-b\delta}{b\delta} \cdot \frac{1}{a}$. The second cannot be dropped. With two periods, $a = 2$, $\delta = \frac{1}{2}$, $v \sim U(0,1)$, shocks uniform on $[0, \frac{1}{2}]$, and $\phi(v) = v/4$ below 0.49, $3v/4 - 1/4$ above 0.51 and quadratic in between (so $\phi'' = 25$ on that interval), the two incentive constraints between types 0.49 and 0.51 sum to $-1/2500 < 0$, so no transfers implement the rule.

5 Why it matters

PST's bandit auction is the standard illustration of how optimal dynamic mechanisms handle buyers who learn by consuming, and equation (15) is the formula a reader would implement. Taken literally, it undervalues experimentation. By averaging along each path, it caps the value of continuing after good news at one half in the example, so it refuses trials that the optimal mechanism makes. The fix is the textbook Gittins index with constant costs, under which PST's argument goes through. The envelope formula, revenue equivalence and the implementability theorems are untouched.

Notes. All page numbers refer to the published article. We found no published correction, but the search was not exhaustive. The cutoff solves $g(r^*) = 0$ with $r^* = -0.27603$, and the alternative mechanism earns $\frac{1}{4}[1 + 2\Phi(1) + \phi(1) - \Phi(r^*)]$.

References. Gittins, J., K. Glazebrook and R. Weber (2011), *Multi-Armed Bandit Allocation Indices*, 2nd ed., Wiley. Pavan, A., I. Segal and J. Toikka (2014), Dynamic mechanism design: a Myersonian approach, *Econometrica* 82(2), 601–653, and Supplemental Material.